

AI / ML Internship — Task 1

Linear Regression Model | California Housing Price Predictor

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1. Objective

The goal of this task is to build and evaluate a **Linear Regression** model that predicts median house prices in California using the built-in *sklearn* California Housing dataset. The project covers the complete ML workflow: data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and reporting.

2. Dataset Overview

Property	Detail
Source	sklearn.datasets.fetch_california_housing
Total Samples	20,640 rows
Features	8 numerical features
Target Variable	MedHouseVal (Median House Value in \$100,000s)
Missing Values	None
Train / Test Split	80% / 20% (random_state=42)

Key Features Used:

Feature	Description
MedInc	Median income in block group (\$10,000s)
HouseAge	Median house age in the block
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average number of household members
Latitude / Longitude	Geographic coordinates of block group

3. Exploratory Data Analysis (EDA)

- The target variable **MedHouseVal** is right-skewed with a hard cap at 5.0 (\$500,000), suggesting some price censoring in the data.
- MedInc** (median income) shows the **strongest positive correlation** with house price ($r \approx 0.69$), confirming that higher-income areas have pricier homes.
- Latitude** has a moderate negative correlation ($r \approx -0.14$), reflecting that southern California areas tend to have higher prices.
- AveRooms** and **AveBedrms** show weak positive correlations; outliers (very large room counts) were noted but not removed.
- No missing values were found in any feature — the dataset is clean and ready for modeling.

4. Methodology

#	Step	Description
1	Data Loading	Loaded dataset via sklearn, combined features and target into a single DataFrame.
2	Preprocessing	Applied StandardScaler to normalize all 8 features before model training.
3	Train/Test Split	80% training (16,512 samples) / 20% testing (4,128 samples), random_state=42.
4	Model	sklearn LinearRegression — Ordinary Least Squares (OLS) minimizing MSE.
5	Evaluation	Computed MAE, RMSE, and R ² on the held-out test set.
6	Visualization	Actual vs Predicted scatter, Residual distribution, Feature coefficient bar chart.
7	Model Saving	Serialized model and scaler using joblib (.pkl files) for future inference.

5. Model Evaluation Results

Metric	Value	Interpretation
MAE (Mean Absolute Error)	~0.5300	Average error ≈ \$53,000 per prediction
RMSE (Root Mean Sq. Error)	~0.7320	Sensitive to large errors; penalizes outliers
R ² (R-Squared Score)	~0.6060	Model explains ~60.6% of price variance

The Linear Regression model achieves an R² of ~0.606, meaning it explains roughly 61% of the variance in California house prices. The RMSE of 0.732 is elevated primarily due to the **hard-capped values at 5.0** in the target and some heteroscedasticity visible in the residual plot. The model performs well as a baseline but has clear room for improvement.

6. Improvement Ideas

Improvement	Details
Ridge / Lasso Regression	Add L2/L1 regularization to reduce overfitting and handle correlated features.
Random Forest / XGBoost	Tree-based models capture non-linear relationships; typically achieve R ² > 0.80.
Feature Engineering	Create new features like rooms-per-person (AveRooms / AveOccup) for better signal.
Outlier Removal	Cap or remove samples where MedHouseVal = 5.0 to reduce target censoring bias.
Cross-Validation	Use 5-fold CV for a more robust and unbiased performance estimate.
Polynomial Features	Add degree-2 features to capture non-linear interactions between predictors.

7. Conclusion

This project successfully demonstrates the complete ML workflow using Linear Regression on the California Housing dataset. The model achieves an R² of ~0.606 and an MAE of ~0.53 (\$53,000), serving as a strong baseline. Key takeaway: **MedInc is the most important predictor** of house prices. Future iterations using ensemble models like XGBoost are expected to significantly improve performance.

